

QUESTION 1

Implement calculator using switch case

CODE:

```
#include <stdio.h>

int main() {
    float a, b;
    char op;

    printf("Enter expression (e.g. 5 + 3): ");
    scanf("%f %c %f", &a, &op, &b);

    switch(op) {
        case '+':
            printf("Result = %.2f\n", a + b);
            break;
        case '-':
            printf("Result = %.2f\n", a - b);
            break;
        case '*':
            printf("Result = %.2f\n", a * b);
            break;
        case '/':
            if(b != 0)
                printf("Result = %.2f\n", a / b);
            else
                printf("Division by zero not allowed.\n");
            break;
        default:
            printf("Invalid operator.\n");
    }

    return 0;
}
```

OUTPUT:

```
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc lab1.c -o lab1
PS C:\Users\User1\Desktop\phe_lab\LAB1> .\lab1.exe
Enter expression (e.g. 5 + 3): 3 + 9
Result = 12.00
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S lab1.c
```

C lab1.c
lab1.exe
ASM lab1.s
M Makefile

```
.file "lab1.c"
.text
.section .rdata,"dr"
.align 8
.LC0:
.ascii "Enter expression (e.g. 5 + 3): \0"
.LC1:
.ascii "%f %c %f\0"
.LC2:
.ascii "Result = %.2f\12\0"
.LC4:
.ascii "Division by zero not allowed.\0"
.LC5:
.ascii "Invalid operator.\0"
.text
.globl main
.def main; .scl 2; .type 32; .endef
.seh_proc main
main:
    pushq %rbp
```

```
CC = gcc
CFLAGS = -Wall
TARGET = lab1
SRC = lab1.c
```

```
✓ $(TARGET): $(SRC)
    $(CC) $(CFLAGS) -o $@ $^
```

QUESTION 1

Implement factorial using loop

CODE:

```
#include <stdio.h>

int main() {
    int n, i;
    long long fact = 1;

    printf("Enter a number: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        fact *= i;
    }

    printf("Factorial = %lld\n", fact);

    return 0;
}
```

OUTPUT:

```
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc fac.c -o fac
PS C:\Users\User1\Desktop\phe_lab\LAB1> .\fac.exe
Enter a number: 7
Factorial = 5040

PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S fac.c
```

```

.section .rdata,"dr"
.LC0:
.ascii "Enter a number: \0"
.LC1:
.ascii "%d\0"
.LC2:
.ascii "Factorial = %lld\12\0"
.text
.globl main
.def main; .scl 2; .type 32; .endef
.seh_proc main
main:
    pushq %rbp
    .seh_pushreg %rbp
    movq %rsp, %rbp
    .seh_setframe %rbp, 0
    subq $64, %rsp
    .seh_stackalloc 64
    .seh_endprologue
    call main

```

C fac.c

≡ fac.exe

ASM fac.s

QUESTION 3

Implement Fibonacci series using loop

CODE:

```

#include <stdio.h>

int main() {
    int n, i;
    int a = 0, b = 1, c;

    printf("Enter number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series: ");

    for(i = 1; i <= n; i++) {
        printf("%d ", a);
    }
}

```

```

        c = a + b;
        a = b;
        b = c;
    }

    return 0;
}

```

OUTPUT:

```

● PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc fibonacci.c -o fibonacci
● PS C:\Users\User1\Desktop\phe_lab\LAB1> .\fibonacci.exe
Enter number of terms: 3
Fibonacci Series: 0 1 1
● PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S fibonacci.c

```

C fibonacci.c
≡ fibonacci.exe
ASM fibonacci.s

```

.file "fibonacci.c"
.text
.section .rdata,"dr"
.LC0:
.ascii "Enter number of terms: \0"
.LC1:
.ascii "%d\0"
.LC2:
.ascii "Fibonacci Series: \0"
.LC3:
.ascii "%d \0"
.text
.globl main
.def main; .scl 2; .type 32; .endef
.seh_proc main
main:

```

QUESTION 4

Implement a program to detect prime number

CODE:

```
#include <stdio.h>
```

```
int main() {
    int n, i, prime = 1;

    printf("Enter a number: ");
    scanf("%d", &n);

    if(n <= 1)
        prime = 0;
    else {
        for(i = 2; i <= n / 2; i++) {
            if(n % i == 0) {
                prime = 0;
                break;
            }
        }
    }

    if(prime)
        printf("%d is Prime.\n", n);
    else
        printf("%d is Not Prime.\n", n);

    return 0;
}
```

OUTPUT:

```

PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc prime.c -o prime
PS C:\Users\User1\Desktop\phe_lab\LAB1> .\prime.exe
Enter a number: 6
6 is Not Prime.
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S prime.c

```

M Makefile
C prime.c
 prime.exe
ASM prime.s

```

.file "prime.c"
.text
.section .rdata,"dr"
.LC0:
.ascii "Enter a number: \0"
.LC1:
.ascii "%d\0"
.LC2:
.ascii "%d is Prime.\12\0"
.LC3:
.ascii "%d is Not Prime.\12\0"
.text
.globl main
.def main; .scl 2; .type 32; .endef
.seh_proc main
main:
pushq %rbp

```

QUESTION 5

Implement a program to count digits in a number

CODE:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, count = 0;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &n);
```

```
    while(n != 0) {
```

```

        count++;
        n /= 10;
    }

    printf("Number of digits = %d\n", count);

    return 0;
}

```

OUTPUT:

```

PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc count.c -o count
PS C:\Users\User1\Desktop\phe_lab\LAB1> .\count.exe
Enter a number: 7
Number of digits = 1
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S count.c

```

```

C count.c
count.exe
ASM counts

```

```

.file "count.c"
.text
.section .rdata,"dr"
.LC0:
.ascii "Enter a number: \0"
.LC1:
.ascii "%d\0"
.LC2:
.ascii "Number of digits = %d\12\0"
.text
.globl main
.def main; .scl 2; .type 32; .endef
.seh_proc main
main:
    pushq %rbp
    .seh pushreg %rbp

```

QUESTION 6

Implement a program to reverse the number

CODE:

```
#include <stdio.h>
```

```

int main() {
    int n, reverse = 0, rem;

    printf("Enter a number: ");
    scanf("%d", &n);

    while(n != 0) {
        rem = n % 10;
        reverse = reverse * 10 + rem;
        n /= 10;
    }

    printf("Reversed Number = %d\n", reverse);

    return 0;
}

```

OUTPUT:

```

PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc reverse.c -o reverse
PS C:\Users\User1\Desktop\phe_lab\LAB1> .\reverse.exe
Enter a number: 786
Reversed Number = 687
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S reverse.c
PS C:\Users\User1\Desktop\phe_lab\LAB1>

```

C reverse.c
 reverse.exe
 ASM reverse.s

```

.file    "reverse.c"
.text
.section .rdata,"dr"
.LC0:
.ascii  "Enter a number: \0"
.LC1:
.ascii  "%d\0"
.LC2:
.ascii  "Reversed Number = %d\12\0"
.text
.globl  main
.def    main; .scl    2; .type    32; .endef
.seh_proc  main
main:
    pushq   %rbp
    .seh_pushreg   %rbp
    movq    %rsp, %rbp
    .seh_setframe  %rbp, 0
    subq    $48, %rsp
    .seh_stackalloc 48

```

QUESTION 7

Implement separate functions to insert, delete, and search element in an array. Implement a main passing user provided array as input to these functions.

CODE:

```

#include <stdio.h>

void insert(int arr[], int *size, int pos, int value) {
    int i;

    for(i = *size; i > pos; i--)
        arr[i] = arr[i - 1];

    arr[pos] = value;
    (*size)++;
}

void deleteElement(int arr[], int *size, int pos) {
    int i;

```

```

        for(i = pos; i < *size - 1; i++)
            arr[i] = arr[i + 1];

        (*size)--;
    }

int search(int arr[], int size, int value) {
    int i;

    for(i = 0; i < size; i++) {
        if(arr[i] == value)
            return i;
    }

    return -1;
}

int main() {
    int arr[100], size, i;
    int pos, value;

    printf("Enter size: ");
    scanf("%d", &size);

    printf("Enter elements:\n");
    for(i = 0; i < size; i++)
        scanf("%d", &arr[i]);

    printf("Enter position and value to insert: ");
    scanf("%d%d", &pos, &value);
    insert(arr, &size, pos, value);

    printf("Array after insertion:\n");
    for(i = 0; i < size; i++)
        printf("%d ", arr[i]);

    printf("\nEnter position to delete: ");
    scanf("%d", &pos);
    deleteElement(arr, &size, pos);

    printf("Array after deletion:\n");
    for(i = 0; i < size; i++)
        printf("%d ", arr[i]);

```

```

printf("\nEnter value to search: ");
scanf("%d", &value);

int result = search(arr, size, value);

if(result != -1)
    printf("Found at index %d\n", result);
else
    printf("Not Found\n");

return 0;
}

```

OUTPUT:

```

● PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc arrfunc.c -o arrfunc
● PS C:\Users\User1\Desktop\phe_lab\LAB1> .\arrfunc.exe
Enter size: 4
Enter elements:
2
3
4
5
Enter position and value to insert: 2
9
Array after insertion:
2 3 9 4 5
Enter position to delete: 1
Array after deletion:
2 9 4 5
Enter value to search: 5
Found at index 3
● PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S arrfunc.c
● PS C:\Users\User1\Desktop\phe_lab\LAB1>

```

 arrfunc.c
 arrfunc.exe
 arrfunc.s

```

.file    "arrfunc.c"
.text
.globl  insert
.def    insert; .scl    2; .type    32; .endef
.seh_proc  insert
insert:
    pushq    %rbp
    .seh_pushreg    %rbp
    movq    %rsp, %rbp
    .seh_setframe    %rbp, 0
    subq    $16, %rsp
    .seh_stackalloc 16
    .seh_endprologue
    movq    %rcx, 16(%rbp)
    movq    %rdx, 24(%rbp)
    movl    %r8d, 32(%rbp)
    movl    %r9d, 40(%rbp)

```

QUESTION 8

Implement a program to do string reverse, compare two strings, concatenate strings

CODE:

```

#include <stdio.h>
#include <string.h>

int main() {
    char str1[100], str2[100], rev[100];
    int i, len;

    printf("Enter first string: ");
    scanf("%s", str1);

    printf("Enter second string: ");
    scanf("%s", str2);

    len = strlen(str1);

    for(i = 0; i < len; i++)
        rev[i] = str1[len - 1 - i];

    rev[len] = '\0';

```

```
printf("Reversed String: %s\n", rev);

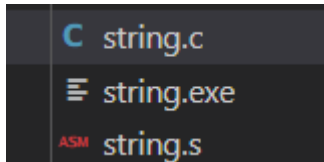
if(strcmp(str1, str2) == 0)
    printf("Strings are equal\n");
else
    printf("Strings are not equal\n");

strcat(str1, str2);
printf("Concatenated String: %s\n", str1);

return 0;
}
```

OUTPUT:

```
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc string.c -o string
PS C:\Users\User1\Desktop\phe_lab\LAB1> .\string.exe
Enter first string: asifa
Enter second string: siraj
Reversed String: afisa
Strings are not equal
Concatenated String: asifasiraj
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S string.c
```



```

.file    "string.c"
.text
.section .rdata,"dr"
.LC0:
.ascii  "Enter first string: \0"
.LC1:
.ascii  "%s\0"
.LC2:
.ascii  "Enter second string: \0"
.LC3:
.ascii  "Reversed String: %s\12\0"
.LC4:
.ascii  "Strings are equal\0"
.LC5:
.ascii  "Strings are not equal\0"
.LC6:
.ascii  "Concatenated String: %s\12\0"
.text
.globl  main

```

QUESTION 9

Implement a program to detect if string is a palindrome.

CODE:

```

#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i, len, flag = 1;

    printf("Enter a string: ");
    scanf("%s", str);

    len = strlen(str);

    for(i = 0; i < len / 2; i++) {
        if(str[i] != str[len - 1 - i]) {
            flag = 0;
            break;
        }
    }
}

```

```

    }

    if(flag)
        printf("Palindrome String\n");
    else
        printf("Not a Palindrome String\n");

    return 0;
}

```

OUTPUT:

- PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc palin.c -o palin
- PS C:\Users\User1\Desktop\phe_lab\LAB1> .\palin.exe
Enter a string: asifasiraj
Not a Palindrome String
- PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S palin.c

C palin.c
≡ palin.exe
ASM palin.s

```

.file    "palin.c"
.text
.section .rdata,"dr"
.LC0:
.ascii  "Enter a string: \0"
.LC1:
.ascii  "%s\0"
.LC2:
.ascii  "Palindrome String\0"
.LC3:
.ascii  "Not a Palindrome String\0"
.text
.globl  main
.def    main; .scl    2; .type  32; .endef
.seh_proc  main
main:
    pushq    %rbp

```

QUESTION 10

Implement a program to convert decimal to octal and hex decimal.

CODE:

```
#include <stdio.h>

int main() {
    int decimal;

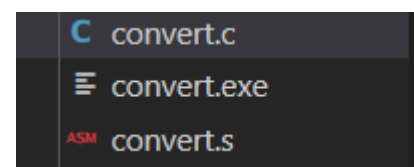
    printf("Enter a decimal number: ");
    scanf("%d", &decimal);

    printf("Octal = %o\n", decimal);
    printf("Hexadecimal = %X\n", decimal);

    return 0;
}
```

OUTPUT:

```
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc convert.c -o convert
PS C:\Users\User1\Desktop\phe_lab\LAB1> .\convert.exe
Enter a decimal number: 7
Octal = 7
Hexadecimal = 7
PS C:\Users\User1\Desktop\phe_lab\LAB1> gcc -S convert.c
```



```
.file "convert.c"
.text
.section .rdata,"dr"
.LC0:
.ascii "Enter a decimal number: \0"
.LC1:
.ascii "%d\0"
.LC2:
.ascii "Octal = %o\12\0"
.LC3:
.ascii "Hexadecimal = %X\12\0"
.text
.globl main
.def main; .scl 2; .type 32; .endef
.seh_proc main
main:
pushq %rbp
```